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Консоль отладки Microsoft Visual Studio
Creating 1000 buses asynchronously...
Finished creating 1000 buses in 180 ms.

--- Demonstrating Synchronization Primitives ---
ManualResetEvent Demo:
Main Thread: Signaling ManualResetEvent in 2 seconds...
Thread 1: Waiting for ManualResetEvent...
Thread 2: Waiting for ManualResetEvent...
Main Thread: ManualResetEvent signaled.
Thread 2: ManualResetEvent signaled! Proceeding.
Thread 1: ManualResetEvent signaled! Proceeding.

AutoResetEvent Demo:
Main Thread: Signaling AutoResetEvent for Thread A in 2 seconds...
Thread B: Waiting for AutoResetEvent...
Thread A: Waiting for AutoResetEvent...
Main Thread: AutoResetEvent signaled for Thread A.
Thread B: AutoResetEvent signaled! Proceeding.
Main Thread: Signaling AutoResetEvent for Thread B in 1 second...
Main Thread: AutoResetEvent signaled for Thread B.
Thread A: AutoResetEvent signaled! Proceeding.

Mutex Demo (Simulating Cross-Process Lock):
Main Thread: Acquired global Mutex. Performing some operation...
Main Thread: Mutex operation complete.
Main Thread: Released global Mutex.

--- CRUD Operations & Data Analysis ---
Saving all buses to file...
Data saved to bus data.json
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Buses saved successfully.
Total buses in service: 1000

--- Statistical Analysis ---
Fuel Consumption (L/100km): Min=15,00, Max=29,98, Avg=22,59
Passenger Capacity: Min=20, Max=99, Avg=59,50
Year: Min=1990, Max=2025, Avg=2006,91

--- Pagination Demo ---
Page 1 (first 10 buses):
Id: e3d83798-0ce7-43fc-bc4b-3942e58a7c23, Model: Bus_Model_1ec9, Year: 2004, Fuel Consumption: 28,12 L/100km, Capacity: 93
Id: 5a29fe68-2406-4a82-93b1-3c08cd26f9ad, Model: Bus_Model_0ae8, Year: 1994, Fuel Consumption: 21,85 L/100km, Capacity: 47
Id: 1aeadd625-0c0b-4314-bfed-2db0b5bcd6b8, Model: Bus_Model_6613, Year: 2019, Fuel Consumption: 20,75 L/100km, Capacity: 47
Id: b5bd6cb5-d569-4fef-bf9e-6a4c23ee6c69, Model: Bus_Model_2d92, Year: 2015, Fuel Consumption: 21,23 L/100km, Capacity: 83
Id: 8820b6bb-175a-4fe4-98b3-3d99f150ea38, Model: Bus_Model_5e28, Year: 2016, Fuel Consumption: 27,1 L/100km, Capacity: 35
Id: 94c996cd-767c-4cb6-b9eb-27419983d090, Model: Bus_Model_04fd, Year: 1999, Fuel Consumption: 21,1 L/100km, Capacity: 75
Id: 423faf24-4357-4110-b53f-475d70897b58, Model: Bus_Model_eca4, Year: 2019, Fuel Consumption: 20,01 L/100km, Capacity: 96
Id: 0653660a-d9bc-4cd8-bc0c-12eb5220f5a2, Model: Bus_Model_d8fc, Year: 1992, Fuel Consumption: 17,3 L/100km, Capacity: 20
Id: df9703a3-d1c7-4613-b1af-d90e4b3bab14, Model: Bus_Model_cf51, Year: 2007, Fuel Consumption: 15,57 L/100km, Capacity: 79
Id: e628c779-5245-4e84-9b0b-9de795c6c5bd, Model: Bus_Model_5c14, Year: 1993, Fuel Consumption: 15,31 L/100km, Capacity: 79
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Page 2 (next 10 buses):
Id: bb807ad1-d714-4bc9-a0c8-98eb45adc059, Model: Bus_Model_b4d2, Year: 1994, Fuel Consumption: 28,46 L/100km, Capacity: 32
Id: 40a6dd8c-b472-48e4-bcbd-2937fd239cf7, Model: Bus_Model_8a84, Year: 2005, Fuel Consumption: 20 L/100km, Capacity: 57
Id: 1193301f-b6e2-4e7e-9124-ae73cd74b756, Model: Bus_Model_df1c, Year: 1993, Fuel Consumption: 25,64 L/100km, Capacity: 21
Id: 30f3c793-05b2-4bce-a3e7-920f1e499e5d, Model: Bus_Model_8455, Year: 2020, Fuel Consumption: 28,09 L/100km, Capacity: 40
Id: 7d8724a1-a883-4dc9-81cb-2f24b91a502e, Model: Bus_Model_7d57, Year: 2013, Fuel Consumption: 29,97 L/100km, Capacity: 32
Id: 78eae8a-362a-4e91-b499-d533a1c3ed1a, Model: Bus_Model_ba77, Year: 2002, Fuel Consumption: 21 L/100km, Capacity: 33
Id: 04da464c-dfcf-4bcf-b85d-2d1ea4d80618, Model: Bus_Model_5ab4, Year: 2004, Fuel Consumption: 28,55 L/100km, Capacity: 27
Id: 4c6e6b16-48a1-42fe-9498-340c3919a8ca, Model: Bus_Model_66f6, Year: 2023, Fuel Consumption: 18,3 L/100km, Capacity: 50
Id: 0724a985-8635-4dea-8c50-6d1f34bb5b70, Model: Bus_Model_503f, Year: 2003, Fuel Consumption: 27,32 L/100km, Capacity: 57
Id: dc5c4a2c-7dc8-45d1-8718-3245abfd3fffb, Model: Bus_Model_6597, Year: 1992, Fuel Consumption: 28,87 L/100km, Capacity: 76

--- Update Demo ---
Original bus: Id: e3d83798-0ce7-43fc-bc4b-3942e58a7c23, Model: Bus_Model_1ec9, Year: 2004, Fuel Consumption: 28,12 L/100km, Capacity: 93
Bus updated successfully: Id: e3d83798-0ce7-43fc-bc4b-3942e58a7c23, Model: Updated Model, Year: 2004, Fuel Consumption: 28,12 L/100km, Capacity: 93

--- Remove Demo ---
Attempting to remove bus: 5a29fe68-2406-4a82-93b1-3c08cd26f9ad

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--- Remove Demo ---
Attempting to remove bus: 5a29fe68-2406-4a82-93b1-3c08cd26f9ad
Bus 5a29fe68-2406-4a82-93b1-3c08cd26f9ad removed successfully.
Total buses after removal: 999
Data saved to bus_data.json

Application finished.
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